

PRODUCTION OF COHERENT STATES IN BIOLOGICAL SYSTEMS

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It is shown that Glauber-type coherent states exist in biological systems as a result of the broken symmetry of the free phonons. The symmetry of the system is reduced by the presence of the interaction terms in the hamiltonian. The coherent states so produced allow for the existence of giant dipoles capable of long-range interactions.

During recent years Fröhlich [1,2] has shown that under certain conditions a phenomenon resembling the well-known Bose—Einstein condensation can occur and produces important biological consequences. One of these consequences is the fact that if the quasiparticles, that have undergone condensation, are characterized by the presence of dipoles then the entire condensate behaves as a giant dipole. If this giant dipole possesses oscillatory character then Fröhlich [3] has shown that long-range (r^{-3}) potentials between two dipoles can occur. Experimentally, Rowlands and co-workers [4] have shown that human red blood cells, in the presence of metabolism and under conditions where transmembrane potentials exist, are capable of exerting long-range forces on each other that result in the production of aggregates commonly referred to as rouleaux. In order to develop a theoretical model for this phenomenon the author of this paper has utilized the idea of a giant dipole on each cell with the concomitant production of a long-range interaction [5,6]. This theory has successfully accounted for Rowlands observation. The theoretical structure is based upon the primary premise regarding the possibility of a dipolar coherent state on each cell. In his original work Fröhlich used a phenomenological rate equation for which a steady-state solution was obtained and recast into a standard Bose—Einstein-type distribution (displaying condensation and coherence) after certain assumptions have been introduced. Such an approach, immediately, became the subject of serious attack by Lifshits [7] and the occurrence of condensation was questioned. Wu and Austin [8] considered the problem from a microscopic point of view and refuted Lifshits contention. The approach adopted by Wu and Austin is in many ways similar to that of Fröhlich since a rate equation is first obtained from the microscopic hamiltonian and then a steady-state solution resembling that of Fröhlich is derived. This derivation has not, however, settled the matter since Yushina [9] repeated the calculations using essentially the Wu—Austin hamiltonian and showed that the specific and desired form of the steady-state solution, leading to Bose—Einstein condensation, does not result. The work of Yushina has been recently attacked by Mills [10–12] who has proved the existence of non-thermal excitations even under conditions where Bose condensation fails. The work presented by Mills provides a firm theoretical foundation for phenomena requiring the existence of polarization waves.

In this paper we adopt the Wu—Austin hamiltonian and proceed via an alternative route that does not depend upon a rate equation. Our approach is a group-theoretic one and is based upon the fact that the interaction terms of the hamiltonian break the phase symmetry of the free hamiltonian. This condition of broken symmetry leads to a translation of the phonon-field operators by a c-number and the production of a coherent state. Since the operators of the phonons of interest are dipolar in nature it follows that the system may be associated with a giant dipole. This resultant dipole is the outcome of the coherent state because a chaotic state would produce a vanishing average value for the dipole moment.

The Wu–Austin hamiltonian is given by

$$H = H_0 + H_1, \quad (1)$$

where

$$H_0 = \sum_1 \epsilon_1^a a_1^\dagger a_1 + \sum_1 \epsilon_1^b b_1^\dagger b_1 + \sum_1 \epsilon_1^p P_1^\dagger P_1, \quad (2)$$

$$H_1 = \sum_{1,2} (\lambda b_1^\dagger a_2 + \lambda^* b_1 a_2^\dagger) + \sum_{1,2,3} (\chi a_1^\dagger a_2 b_3^\dagger + \chi^* a_1 a_2^\dagger b_3) + \sum_{1,2} (\xi P_1 a_2^\dagger + \xi^* P_1^\dagger a_2). \quad (3)$$

Here, the definitions are the same as those used in ref. [8] except the indices are enumerated as numbers and the energies of the phonons are given the symbol ϵ .

If we now make the transformation

$$a_i = e^{i\theta} a_i \quad (4)$$

(where θ is some phase) we find that H_0 is invariant to this transformation but H_1 possesses a lower symmetry. Thus the total hamiltonian H is of lower symmetry than H_0 and we do not expect the quantum statistical average $\langle\langle a_i \rangle\rangle$ to vanish with the exact density operator ρ

$$\rho = e^{-\beta H} / Q, \quad (5)$$

where Q is the exact partition function. In the subsequent analysis we adopt the following definitions

$$\langle \text{operator} \rangle \equiv \text{tr}(e^{-\beta H_0} / Q_0) (\text{operator}), \quad \langle\langle \text{operator} \rangle\rangle \equiv \text{tr}(e^{-\beta H} / Q) (\text{operator}). \quad (6,7)$$

Physically, we may view a non-vanishing $\langle\langle a_i \rangle\rangle$ as follows:

(1) We start with a collection of three types of non-interacting phonons: the system phonons, the bath phonons and the pump phonons.

(2) This collection is in complete thermal equilibrium and yields

$$\langle a_i \rangle = 0 \quad (8)$$

(3) When the interactions are switched on, via H_1 , a translation of the operators occurs. This translation is given by

$$a_i = a_i + c_i, \quad (9)$$

with

$$c_i \equiv \langle\langle a_i \rangle\rangle. \quad (10)$$

The term translation or displacement was used by Glauber [13] because eq. (9) can be induced by a unitary operator

$$D(c_i) \equiv \exp(c_i a_i^\dagger - c_i^* a_i). \quad (11)$$

In this terminology the vacuum state is converted to an eigenstate of the operator a_i :

$$|c_i\rangle = D(c_i)|0\rangle, \quad (12)$$

where

$$a_i |c_i\rangle = c_i |c_i\rangle. \quad (13)$$

States such as $|c_i\rangle$ are called *coherent states*. If we assume that the dipole moment μ_i of the phonons are related to the coordinate q_i

$$\mu_l \propto q_l \propto (a_l^\dagger + a_l) \quad (14)$$

then in the coherent state

$$\langle \mu_l \rangle \neq 0. \quad (15)$$

In this description the macroscopic dipole moments $\langle \mu_l \rangle$ arise quite naturally and the rest of the theoretical arguments, requiring the existence of these dipole moments, can be carried out. The translation phenomenon indicated by eq. (9) has been recognized as a generalized criterion for Bose–Einstein condensation by other workers (see for example Umezawa [14]). The remainder of this paper will be devoted to the explicit computation of c_l .

From eqs. (7) and (6) along with the definition of H we have

$$c_l = Q^{-1} \text{tr} \{ \exp[-\beta(H_0 + H_1)] a_l \} = \langle \mathcal{O}(\beta) a_l \rangle_{\text{connected}}, \quad (16)$$

where the finite temperature expansion of the density operator is utilized and by the retention of only the connected Feynman diagrams the necessity of computing the exact partition function Q is eliminated. This simplification is the result of the well-known Wick theorem (see for example Paul [15]). The operator $\mathcal{O}(\beta)$, appearing in eq. (6), is defined by

$$\mathcal{O}(\beta) = \sum_{n=0}^{\infty} \left(\frac{-1}{\hbar} \right)^n \frac{1}{n!} \int_0^{\beta \hbar} du_1 du_2 \dots du_n T H_1(u_1) H_1(u_2) \dots H_1(u_n). \quad (17)$$

Here T is the well-known Wick time ordering operator and u_i are imaginary time variables that are defined within the limits $0 \leq u_i \leq \beta \hbar$. In order to facilitate a diagrammatic development it is convenient to introduce an imaginary time variable u in the operator in a_l in eq. (16). This allows us to calculate $c_l(u)$ from which c_l can be obtained by setting $u = 0$ at the end of the computation.

If we examine the interaction hamiltonian given in eq. (3) we notice the presence of six vertices which are displayed in fig. 1 and the meaning of arrows are given in the following rules:

- (1) Phonons with operators $a_i, a_i^\dagger$ are denoted by a single arrow and since these phonons are dipolar in nature they will be called “active” phonons.
- (2) Bath phonons ($b_i, b_i^\dagger$) are given two arrows.
- (3) Pump phonons ($P_i, P_i^\dagger$) are given three arrows.
- (4) All diagrams will contain an external condensate point which carries the label l . This point will be designated by a square $\square_l$.
- (5) All zero-order Green's (G_{ij}^a, G_{ij}^b and G_{ij}^p) functions will be indicated by thin lines carrying the appropriate numbers of arrows as indicated by rules (1), (2) and (3).
- (6) All renormalized Green's functions ($\bar{G}_{ij}^a, \bar{G}_{ij}^b$ and $\bar{G}_{ij}^p$) will be indicated by heavy lines with the appropriate numbers of arrows as indicated by rules (1), (2) and (3).

The simplest renormalized diagrams that appear in the present theory are shown in fig. 2. It is our main objective, in this paper, to demonstrate the existence of a nonvanishing field translation given by the topologically dis-

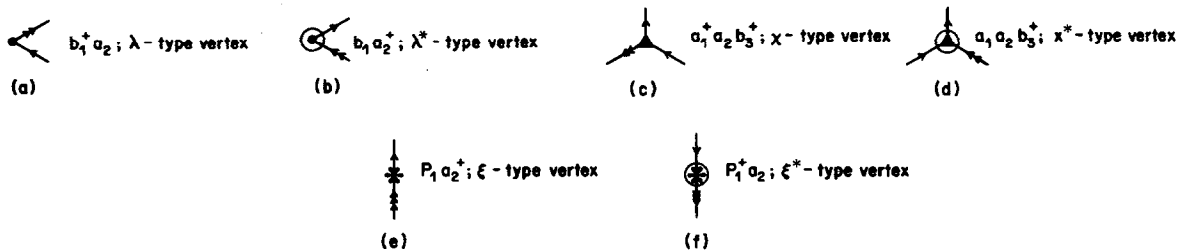


Fig. 1.

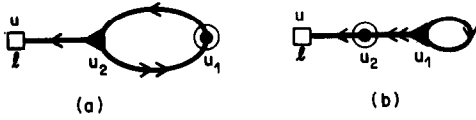


Fig. 2.

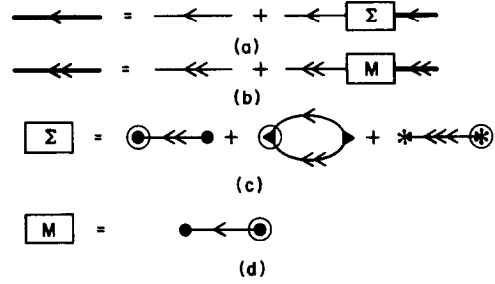


Fig. 3.

tinct diagrams (2a) and (2b). It is sufficient to show a non-vanishing result from either of these diagrams since possible cancellation amongst such topologically distinct diagrams is very unlikely.

For the renormalization procedure we adopt the diagrams displayed in fig. 3, where we use the symbol Σ to designate the self-energies of the active phonons and the symbol M for the self-energies of the bath phonons. The actual computations are done, most conveniently, in a frequency space which is related to the imaginary time space by the following equation:

$$f(u) = \frac{1}{\beta\hbar} \sum_{n=-\infty}^{\infty} f(\omega_n) \exp(-i\omega_n u), \quad (18)$$

where the discrete frequencies ω_n are given for the Bose operators by

$$\omega_n = 2\pi n / \beta\hbar. \quad (19)$$

The zero-order Green's function (quantities corresponding to the thin lines) for the three kinds of phonons are given by

$$G_{ij}^s(\omega_n) = (i\omega_n - \hbar^{-1}\epsilon_i^s)^{-1} \delta_{ij} \quad (s = a, b \text{ or } P). \quad (20)$$

With this definition of the zero-order Green's functions and employing the diagrams of fig. 3 we obtain

$$\bar{G}_{i,j}^a(\omega_m) = \delta_{ij} / [i\omega_m - \hbar^{-1}\epsilon_i^a - \Sigma(\omega_m)], \quad \bar{G}_{i,j}^b(\omega_m) = \delta_{ij} / [i\omega_m - \hbar^{-1}\epsilon_i^b - M(\omega_m)]. \quad (21,22)$$

The self-energies are given by

$$\Sigma(\omega_m) = \frac{|\lambda|^2}{2\hbar^2} \sum_2 \frac{1}{i\omega_m - \hbar^{-1}\epsilon_2^b} - |\chi|^2 \sum_{2,3} \frac{f(\epsilon_2^a) - f(-\epsilon_3^b)}{i\omega_m + \hbar^{-1}(\epsilon_2^a + \epsilon_3^b)} + |\xi|^2 \sum_2 \frac{1}{i\omega_m - \hbar^{-1}\epsilon_2^P}, \quad (23)$$

$$M(\omega_m) = \frac{|\lambda|^2}{2\hbar^2} \sum_2 \frac{1}{i\omega_m - \hbar^{-1}\epsilon_2^a}, \quad f(\epsilon_d^s) \equiv [\exp(\beta\epsilon_d^s) - 1]^{-1}. \quad (24,25)$$

If eqs. (21) and (22) are substituted for the renormalized Green's functions in fig. 2b then the field displacement in frequency space becomes

$$c_l(\omega_m) = \frac{\chi\lambda^*}{2\hbar^2 [\hbar^{-1}\epsilon_l^a + \Sigma(0)]} \sum_{2,4} \frac{1}{i\omega_m - \hbar^{-1}\epsilon_2^a - \Sigma(\omega_m)} \frac{1}{i\omega_m - \hbar^{-1}\epsilon_4^b - M(\omega_m)}. \quad (26)$$

From this equation the static-field displacement can be obtained by first transforming into an imaginary time space v and then allowing $u \rightarrow 0$:

$$c_l = \lim_{u \rightarrow 0} \frac{1}{\beta \hbar} \sum_m c_l(\omega_m) \exp(-i\omega_m u) \\ = \frac{\chi \lambda^*}{2\beta \hbar^3 (\hbar^{-1} \epsilon_l^a + \Sigma)} \sum_{2,4,m} \frac{1}{i\omega_m - \hbar^{-1} \epsilon_2^a - \Sigma(\omega_m)} \frac{1}{i\omega_m - \hbar^{-1} \epsilon_4^b - M(\omega_m)}. \quad (27)$$

Here $\Sigma \equiv \Sigma(0)$. The frequency summation over m can be carried out by integrating in the complex plane if the frequency dependence of the self-energies $\Sigma(\omega_m)$ and $M(\omega_m)$ can be neglected. To carry out this process we define the spectral functions for the two self-energies appearing in eq. (27) as follows:

$$\Sigma(z) = \frac{1}{\pi} \int_{-\infty}^{\infty} d\omega' \frac{\Gamma_a(\omega')}{\omega' - z}, \quad M(z) = \frac{1}{\pi} \int_{-\infty}^{\infty} d\omega' \frac{\Gamma_b(\omega')}{\omega' - z}. \quad (28,29)$$

Here, Γ_a and Γ_b are the spectral functions for the two self-energies and in ordinary spectroscopy these quantities play the role of spectroscopic line broadening. The variable z is a complex frequency that represents an analytic continuation of the discrete frequencies ω_m over the entire complex plane. Just above and below the real frequency axes we obtain

$$\Sigma(\omega \pm i\eta) = \frac{1}{\pi} \int_{-\infty}^{\infty} d\omega' \frac{\Gamma_a(\omega')}{\omega' - \omega \mp i\eta} = P \frac{1}{\pi} \int_{-\infty}^{\infty} d\omega' \frac{\Gamma_a(\omega')}{\omega' - \omega} \pm i\Gamma_a(\omega) \equiv S_a(\omega) \pm i\Gamma_a(\omega). \quad (30)$$

Similarly

$$M(\omega \pm i\eta) \equiv S_b(\omega) \pm i\Gamma_b(\omega). \quad (31)$$

Here ω is a real frequency and will be treated as a constant in the frequency summation involved in eq. (27). The actual computation of the function $S_s(\omega)$ and $\Gamma_s(\omega)$ (with $s = a, b$) involves summations shown in eqs. (24) and (25). These summations over the phonon states cannot be carried out exactly and require numerical work. In this paper we do not propose to give actual numerical values for the field translations. It is our purpose to show that such a translation does occur and to display its mathematical form, therefore the actual computation of the frequency shift (S_s) and broadening will not be carried out. If we define the complex frequencies of the dressed phonons by ν_i^a and ν_j^b where

$$\nu_i^a \equiv \hbar^{-1} \epsilon_i^a + S_a \pm i\Gamma_a, \quad \nu_j^b \equiv \hbar^{-1} \epsilon_j^b + S_b \pm i\Gamma_b, \quad (32,33)$$

then the explicit summation over m in eq. (27) yields

$$c_l = \frac{\chi \lambda^*}{2\hbar^2 \nu_1^a} \sum_{2,4} \frac{f(\hbar \nu_2^a) - f(\hbar \nu_4^b)}{\nu_4^b - \nu_2^a}. \quad (34)$$

This equation shows that as long as the coupling constants χ and λ^* are present there will always be a finite field translation. The magnitude of this translation is inversely proportional to the frequency of the dressed mode l ($c_l \propto 1/\nu_l^a$), this implies that the low-energy modes undergo a relatively more severe distortion or the main contribution to the condensate density arises from small values of l . If we compare our result given by eq. (34) with the result of Mills then at a superficial level they appear contradictory. This is not, however, the case. The main difference arises from the fact that eq. (34) does not explicitly contain the coupling constant ξ (or ξ^*) corresponding to the pump and in Mills work it is precisely the energy pump that produces the non-thermal excitations. Moreover the importance of the pump has been well established by various experimental measurements. The apparent contradic-

tion disappears when it is realized that the present result is a group-theoretical one and is a direct consequence of the symmetry properties of the hamiltonian given by eq. (2). It is easy to see that part of the symmetry breaking is induced in the hamiltonian by the linear coupling with the bath through the $\lambda(\lambda^*)$ dependent terms and the form of these terms is the same as that given by the pump-dependent terms. From a purely mathematical standpoint the inclusions of the symmetry-breaking terms in the hamiltonian have already biased the problems towards coherence. In the treatment based upon the rate equation further physical assumptions (not contained in the hamiltonian or the Heisenberg's equations of motion) are introduced and these give the pump the special significance it finally achieves. These additional physical assumptions are perfectly reasonable and the results obtained are correct, however, the work of this paper is meant to augment the phenomenon of coherence given by Mills and not to negate it.

If the index l is taken to be the phonon propagation vector ($\mathbf{k}$) then, in our theory, $\mathbf{k}$ is the propagation vector of the dressed phonons and the condensate will be composed, predominantly, of modes with $|\mathbf{k}| \rightarrow 0$. With this description of the index l we may define the field operators:

$$\psi(\mathbf{r}) = \sum_{\mathbf{k}} a_{\mathbf{k}} \exp(i\mathbf{k} \cdot \mathbf{r}) + \sum_{\mathbf{k}} c_{\mathbf{k}} \exp(i\mathbf{k} \cdot \mathbf{r}). \quad (35)$$

The density matrix becomes

$$\begin{aligned} \rho(\mathbf{r}, \mathbf{r}') &= \langle \psi^\dagger(\mathbf{r}) \psi(\mathbf{r}') \rangle = \sum_{\mathbf{k}\mathbf{k}'} \langle a_{\mathbf{k}}^\dagger a_{\mathbf{k}'} \rangle \exp(-i\mathbf{k} \cdot \mathbf{r} + i\mathbf{k}' \cdot \mathbf{r}') + \sum_{\mathbf{k}\mathbf{k}'} c_{\mathbf{k}}^* c_{\mathbf{k}'} \exp(-i\mathbf{k} \cdot \mathbf{r} + i\mathbf{k}' \cdot \mathbf{r}') \\ &= \sum_{\mathbf{k}} n_{\mathbf{k}} \exp[i\mathbf{k} \cdot (\mathbf{r} - \mathbf{r}')] + \sum_{\mathbf{k}\mathbf{k}'} c_{\mathbf{k}}^* c_{\mathbf{k}'} \exp(-i\mathbf{k} \cdot \mathbf{r} + i\mathbf{k}' \cdot \mathbf{r}'). \end{aligned} \quad (36)$$

The first term of this expression describes the normal (uncondensed) phonons in the system and vanishes when $|\mathbf{r} - \mathbf{r}'| \rightarrow \infty$ due to the oscillatory nature of the exponential and produces no long-range order. The second term clearly does not display this feature and corresponds to the well-known phenomenon of off-diagonal long-range order (ODLRO) that has been observed in other branches of physics [16].

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